

Communication: Language: easy to understand, answers question, follows instructions, concise, precise Chemistry: proper vocab, notation, spacing, logical Math: organized, units, sig digs, math notation, concluding statement, spacing	Overall: vague/basic/OK/good/clearly written
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SCH4U

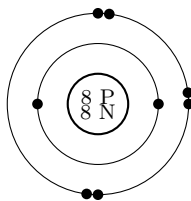
Name: _____

Monday, March 1, 2025

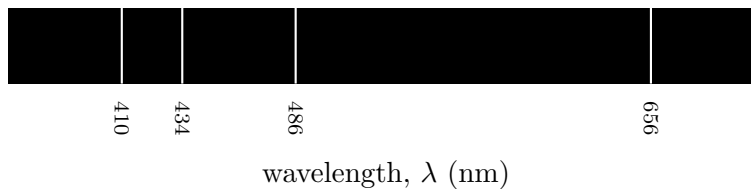
Unit 1: Structures & Properties Test

Total: /35	Comm: /10
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1. [2 marks] What is wrong with the following Bohr-Rutherford diagram with regards to Quantum Mechanics? Be specific.



2. [3 marks] Below is the emission spectra for hydrogen when an electron falls to $n = 2$. Where would an emission line for hydrogen be if an electron falls from $n = 5$ to $n = 3$? Indicate the location of this new emission line relative to the other emission lines. Is this energy transition high energy or low energy? Explain with or without a diagram. Calculations are not needed to answer this question.



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- [6 marks] Assuming there is enough energy to do so, write out the full/longhand electron configuration for the most stable form of Ru, Ru^{3+} & Ru^{6+} . Rank the three electron configurations in order of stability (1 = most stable). Ru is element 44.
- [2 marks] Is the most stable electron configuration for an atom or ion always anomalous? Explain & provide an example to support your answer.
- [4 marks] Predict two realistic ionic charges for Fl (element 114). Write the shorthand electron configuration only. Justify your answer.

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10. [2 marks] What is the same & what is different between the following: AB_2E & AB_2E_2 ? Be precise.
11. [2 marks] Why are ionic solids brittle? What is unique about the structure that allows it to do so? Provide an example of an ionic solid.
12. [2 marks] Explain the humour behind the meme. Be sure to explain/describe the theory needed in order to fully appreciate the chemistry humour in the meme.

